

3.2P: Answer Sheet

Recall task 2.2P *Counter Class* and answer the following questions.

1. How many *Counter* objects were created?

2

2. Variables declared without the **new** keyword are different to the objects created using **new**. In the **Main** function, what is the relationship between the variables initialized with and without the **new** keyword?

The new indicates that an object is being created in relation to a class while one without does not have a class

3. In the **Main** function, explain why the statement **myCounters[2].Reset()**; also changes the value of **myCounters[0]**.

Because **myCounters[2] = myCounters[0]** making one effect the other

4. The difference between *heap* and *stack* is that heap holds “*dynamically allocated memory*.” What does this mean? In your answer, focus on the size and lifetime of the allocations.

The stack is used for storing local variables and function parameters and the memory is allocated at compile time and is automatically unallocated from memory while the heap is allocated during the run time of the program and it stores objects must be manually unallocated from memory

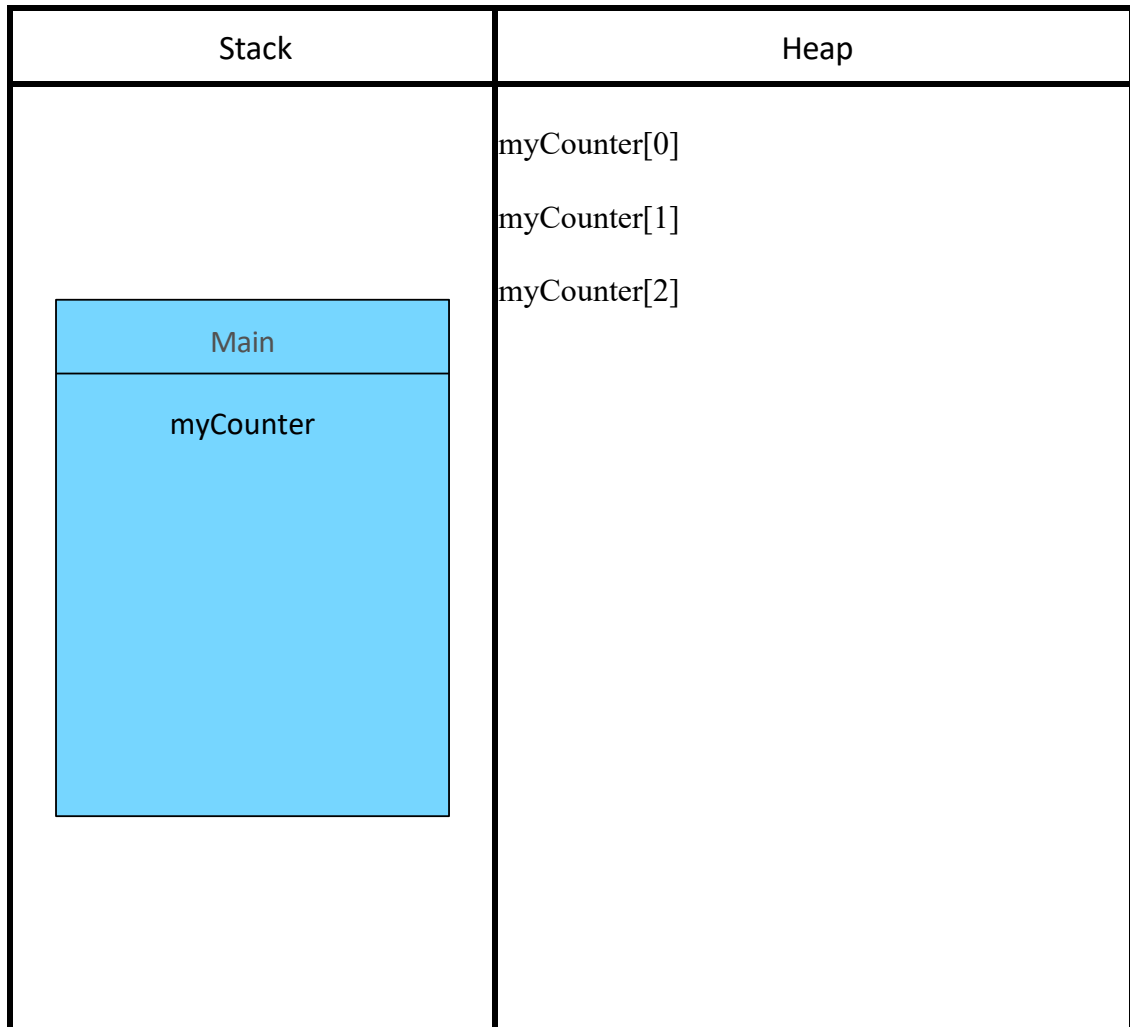
5. Are objects allocated on the heap or on the stack? What about local variables?
The stack is used for local variables while the heap is used for objects

6. What is the meaning of the expression ***new ClassName()***, where *ClassName* refers a class in your application? What is the value of this expression?
you are creating a new object of the class *ClassName()*

7. Consider the statement “*Counter myCounter;*”. What is the value of ***myCounter*** after this statement? Why?

Nothing or null because you create an object but you don't assign it anything

8. Based on the code you wrote in task 2.2P *Counter Class*, draw a diagram showing the locations of the variables and objects in function **Main** and their relationships to one another.



9. If the variable `myCounters` is assigned to `null`, then you want to change the value of `myCounters[X]`, where `X` is the last digit of your student ID, what will happen? Please provide your observation with screenshots and explanation.

The is a no reference for it because you taking away the reference to the object by assigning the variable nothing

Hint. You may want to read this material for this task

<https://learn.microsoft.com/en-us/dotnet/csharp/language-reference/keywords/null>

For further reading at your own.

- Null pointer CrowdStrike Bug, <https://www.thestack.technology/crowstrike-null-pointer-blamed-rca/>
- CrowdStrike Blog, <https://www.crowdstrike.com/blog/tech-analysis-channel-file-may-contain-null-bytes/>